CHAPTER

Introduction to biological databases

2

2.1 Introduction

A database is a large collection of well-organized data designed to remain intact over time. Databases are typically associated with a piece of computer software that allows users to alter (i.e., update), search for, and retrieve specific sets of data that are kept within a computer-based framework. The DBMS software, which stands for "database management system," collaborates with the user application, and the database to store, manage, and change data in such a way that it may be utilized to get information that is of use. To obtain data, and create tables or objects, insertion, deletion, and modification of data in a database, are done through queries written in query language specific to a DBMS. Some of the examples of DBMS include MySQL, Oracle, and PostgreSQL.

Among different sorts of databases, the ones comprising the databases applicable to natural sciences like molecular science and bioinformatics are called biological databases. In the current situation, the significance of biological databases can be perceived from the accompanying focuses:

- Huge volumes of raw data, such as raw sequencing, proteomes, and other types of data, are being generated at a very rapid pace due to many technological advancements in molecular research and proteomics and the cheap cost of high-throughput genome sequencing. As a result, the capability and handling of this incredible data are among the most pressing issues facing genomics at the moment.
- It contributes to the development of customized medicine by helping prescribe the most effective drug.
- This approach allows for the modification of either the patient's or the person's DNA to treat genetic diseases.
- As of now, biological databases have become the focal point of bioinformatics. Through the different information mining devices, all-natural data can be effortlessly gotten to, consequently saving time, assets, and endeavors.
- Additional uses include the manufacture of bioweapons, the investigation of evolutionary processes, the improvement of agricultural yields, and the increase of the nutritional value of food.

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- With the emergence of the machine learning era, NLP is being utilized for automated curation of data, and thus biological databases allow knowledge discovery (when the raw data is entered, there is hidden information that is not known beforehand; entry into the database helps uncover those undiscovered connections between a piece of information entered, which is called knowledge discovery). This facilitates the uncovering of new biological insights from raw data.
- Multiple-point access is possible, as is convenient retrieval of publicly accessible data.
- improved indexing (for global access).
- reduced redundancy as data is curated computationally and manually.

2.1.1 Characteristics of biological data

The majority of other forms of data are substantially simpler in comparison to the complexity of biological data. The definitions of this kind of data have to be able to show both the intricate substructures of the data and the relationships that exist between them to guarantee that no information is lost during the process of modeling biological data. This is the only way to prevent any information from being overlooked. Not just in a hierarchical, binary, or tabular layout, but the data model should be able to display any amount of complexity included in any data schema, relationship, or schema substructure. For instance, the National Center for Biotechnology Information (NCBI) biological data model interprets a biological sequence as a straightforward integer coordinate system, to which many types of data may be connected. The coordinate system is connected to a vast amount of information, including the sequence in which amino acids are found in the body.

The quantity and range of data are quite different from one another.Therefore, there must be some degree of adaptability in the handling of data kinds and values. A single piece of data may need to be represented by more than one data type. This is since biological data structures often include exceptions. In addition, the sorts of data that are gathered for various species and genome projects sometimes overlap with one another.

In biological databases, schemas are subject to constant modification. The majority of relational and object database systems do not now support the addition of new fields to the schema. Therefore, nucleotide sequence databases like GenBank will re-release the whole database with new schemas rather than making incremental improvements to the system as required, which may seem invisible to the user.

People who work with biological data often need to review prior versions of the same data. As an example, GenBank makes use of Accession. The version number in the flat file entries helps keep tabs on the protein sequences. In addition to this, the version contains a GI number, which denotes a certain sequence. While the accession number does not change throughout updates, it does get a new GI number if there is a modification to the protein sequence. GenBank is a public database that focuses on nucleotides; however, when nucleotides are translated, proteins are created from the nucleotides (Gligorijević and Pržulj, 2015).

2.2 Types of databases

Most databases can be put into one of three main groups: primary databases, secondary databases, and composite databases.

2.2.1 Primary database

Primary databases, which are also known as archival databases, are made up mostly of datasets that were generated through experiments. These datasets include things like nucleotide and protein sequences as well as information on how macromolecules are assembled. Users can augment this fundamental information by adding functional annotations, references, and linkages to other databases. The material was entered into the primary database by the researchers themselves. As soon as the information is received, it is included in the scientific record and assigned a unique number that is referred to as an "accession number." Types of primary databases include the following:

1. **Primary nucleotide sequence databases**: The three most significant databases that save raw nucleic acid sequences and make them accessible to users are GenBank (Bilofsky and Christian, 1988), EMBL (Kanz et al., 2005), and DDBJ (Okubo et al., 2006). Users may access these sequences via these databases. Users may also access the sequences by navigating through these databases. You may access GenBank in the United States by going to the internet gateway for the National Center for Biotechnology Information. GenBank is a database that stores genetic sequences. The headquarters of GenBank may be found in the United States of America. In their respective regions of the globe, the European Molecular Biology Laboratory (EMBL) and the DNA Databank of Japan (DDJB) may be located. The European Molecular Biology Laboratory, which is also known as EMBL, is in the UK, while the DDJB is in Japan.
2. **Microarray/Functional genomics databases:** The study of trials that make use of high-throughput technologies to assess transcripts, proteins, and metabolites is what is known as the field of functional genomics. These investigations need a great deal more information regarding the design, sample, and procedures utilized than genomics does since they are highly dependent on the circumstances. For functional genomics, we want databases that adhere to both the current and emerging data quality requirements. There are a few functional genomic databases that concentrate on microbial genomes, and there are a few different methods that may be used to build up functional genomic databases. However, functional genomic databases are not yet particularly prevalent.

3. **Protein sequences and structure databases**: PIR-PSD and SWISS-PROT (Boeckmann et al., 2003) are both examples of databases that hold the protein sequences that have been determined. PIR-PSD is located at the National Biomedical Research Foundation (NBRF) in the United States of America, while SWISS-PROT is housed in Switzerland at the Swiss Biotechnology Institute (SBI). During the development of the PIR-PSD, a collaboration between the PIR, the MIPS (Munich Information Center for Protein Sequences, Germany), and the JIPID (Japan International Protein Information Database, Japan) was essential. The PIR-PSD is now a complete object-relational database management system that is well annotated and has no unnecessary components (DBMS). The PIR-PSD is unique in that it organizes protein sequences into groups by using the concept of "superfamilies" as its organizing principle. In addition, the PIR-PSD sequence is partitioned into groups according to the sequence motifs and homology domains that it contains. Homology domains might be compared to the building blocks of evolution, while sequence motifs refer to functional locations or regions that have not undergone any changes. The approach of categorization makes it simpler to see how the sequence, function, and structure are all interconnected with one another.

2.2.2 Secondary database

The original database's information is transferred to a secondary database. A secondary sequence database contains data such as the conserved sequence, signature sequence, and active site residues of protein families discovered by repeated sequence alignment of a set of related proteins. The PDB entries are stored in a secondary structure database. There are listings for all alpha proteins, all beta proteins, and so on, according to how they are created. These also include information about a protein's recurring secondary structural motifs. Secondary databases created by various academics and housed at their labs include SCOP, created at Cambridge University; CATH, created at University College London; PROSITE, created by the Swiss Institute of Bioinformatics; and eMOTIF, created at Stanford (Rother et al., 2005).

1. **Protein families, domains, and structure databases:** Protein databases are now indispensable to the practise of modern biology. The structures, functions, and, most importantly, the sequencing of proteins are the subjects of a significant amount of current research. When investigating a novel protein, the investigation often starts with a search via several databases. By comparing individual proteins or whole protein families, we may learn about the connections that exist between the proteins in a genome or between the proteins of different species. This is far more information than we might get by examining a single protein on its own. InterPro, PROSITE, SCOP, CATH, and the NCBI Conserved Domain Database are all included here (CDD).

2. **Protein sequences and functional information databases:** The functional information consists of a collection of data that the functional head needs to successfully perform and administer the function. Because this knowledge is solely helpful for performing that one function, it cannot be put to use in any other context. This information will be used by a manager to plan and handle the work at hand.
3. **Nucleotide (Genes/Genomes) sequence and annotation databases:** The process of obtaining information on the structure and function of a protein or gene from a raw data set is referred to as "genome annotation." This is accomplished by the use of a variety of mining methods, including analysis, comparison, estimate, and precision. It is essential to annotate the genome since sequencing the genome or DNA results in the creation of sequence information that provides no insight into how the system functions. After the genome has been sequenced, the data must next be annotated to provide further details about the genome's structure and the way it functions, includes NCBI, Ensembl, etc.

2.2.3 Composite database

The necessity to search in many places is eliminated when using a composite database since it integrates the information from many different major databases. The search methodology used by each composite database is based on a separate primary database as well as a unique set of criteria. In the composite database, there are many different ways to search for the information you need. Researchers are granted unrestricted access to the nucleotide and protein databases that are stored on the massive, high-availability, redundant array of computer servers that are maintained by the National Center for Biotechnology Information (NCBI), which is the organization that is responsible for hosting these databases. In addition to this, a link is provided to the Online Mendelian Inheritance in Man Database, which has information on the proteins that may be associated with inherited diseases. There are situations in which a data collection may function as either a main or secondary database. For instance, primary peptide sequences may be easily uploaded to the Uniprot database. In addition, Uniprot can collect protein clusters from primary peptide sequences. It may also have the automated and manual explanations from TrEMBL and SwissProt (Bateman et al., 2017).

2.3 Models of databases

The logical framework that is used for the storage of computational results via DBMS is referred to as the database model. The kind of database model that is used is a major factor in determining how effectively data can be retrieved and stored. Earlier database models consisted primarily of two-dimensional data tables or a single file containing fields and their values. But because data has grown so much in the last few decades, database models have become more complicated and linked.

A few of the database models we'll cover in the next sections are listed below.

- Flat File
- Hierarchic
- Network
- Entity-Relationship
- Relational

At least one presentation is possible within the context of every information base administrative structure. The optimal structure is determined by the consistent connection of the application's data as well as the application's requirements, which include exchange rate (speed), unswerving quality, practicability, adaptability, and cost. The perfect structure is determined by these factors. Even though products can provide support for more than one model, the vast majority of information database management systems are designed to revolve around a certain information model. Any given coherent model may be actualized using a variety of different physical information models. Because the choices that are made have such a large impact on how the program is run, the vast majority of database programming will provide the user with some level of control over the process of tuning the physical execution (Fig. 2.1).

A model is not only a means of organizing information; rather, it also outlines a variety of actions that may be carried out on the data. These activities include: For example, the social model characterizes tasks such as "select (venture) and join." Even though the results of these actions might not be clear in a certain question language, they are still the building blocks on which a question language is made.

2.3.1 Flat file

The most fundamental kind of database is one in which information is kept in the form of a file or a two-dimensional array (table) and is organized into columns, rows, and values. In this architecture, the columns are responsible for defining the various fields, while the rows hold the data of a single record and are linked together by a shared ID. The relational model evolved from its predecessor, the flat file model. The fields and values that are part of GenBank entries can be thought of as an example of a basic flat file paradigm in biological databases.

2.3.2 Hierarchical model

The information included in this database model is structured in a manner that is reminiscent of a tree, and each of the other nodes is linked to the one that serves as the tree's root. The hierarchy begins with the root data and develops in the shape of a tree when more child nodes are added to the nodes that are already there and make up the parent nodes. Every child node is connected to exactly one parent

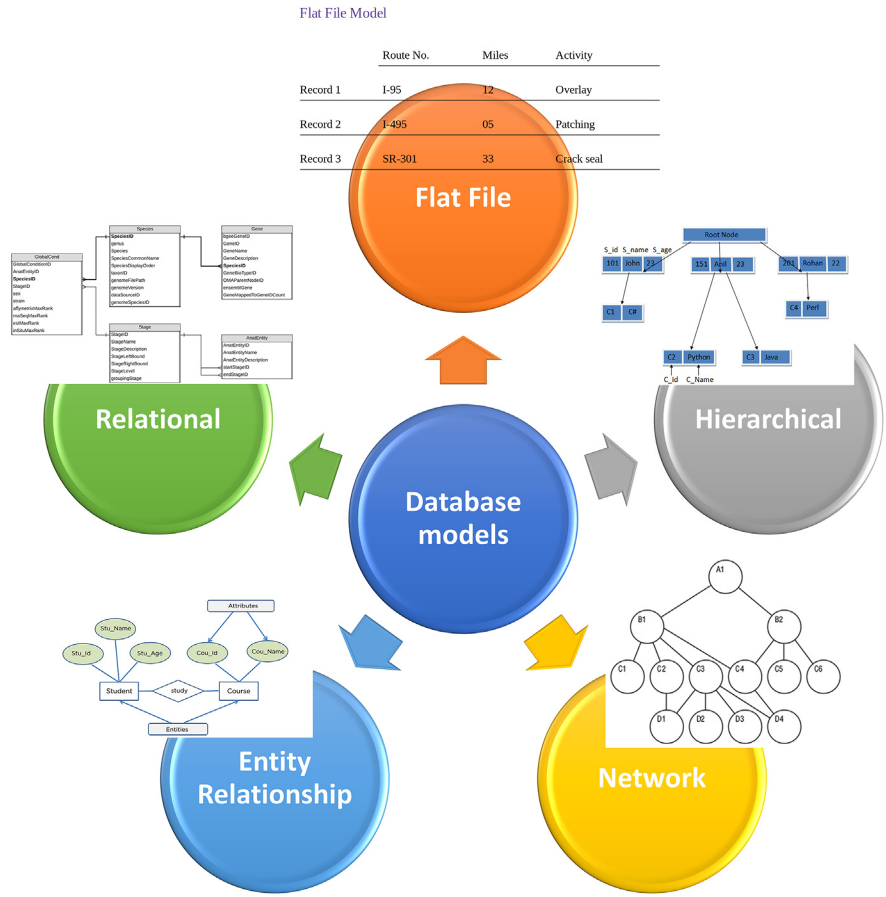

FIGURE 2.1

Some common database models in DBMS.

node. The data in a hierarchical model are laid out in a structure that resembles a tree, and there are relationships between the various categories of data.

Example of a hierarchical database in the biological sciences: the cell type is the root of the biological function network database; each cell has multiple organelles and cytoskeletal elements; each component is involved in multiple pathways and functions; and there can be more levels of hierarchy (Fig. 2.2).

2.3.3 Network model

A network-based extension of the hierarchical model is shown here. The data in this model is structured more like a graph, and a node can have more than one parent node. The data in this database model becomes more interconnected as more

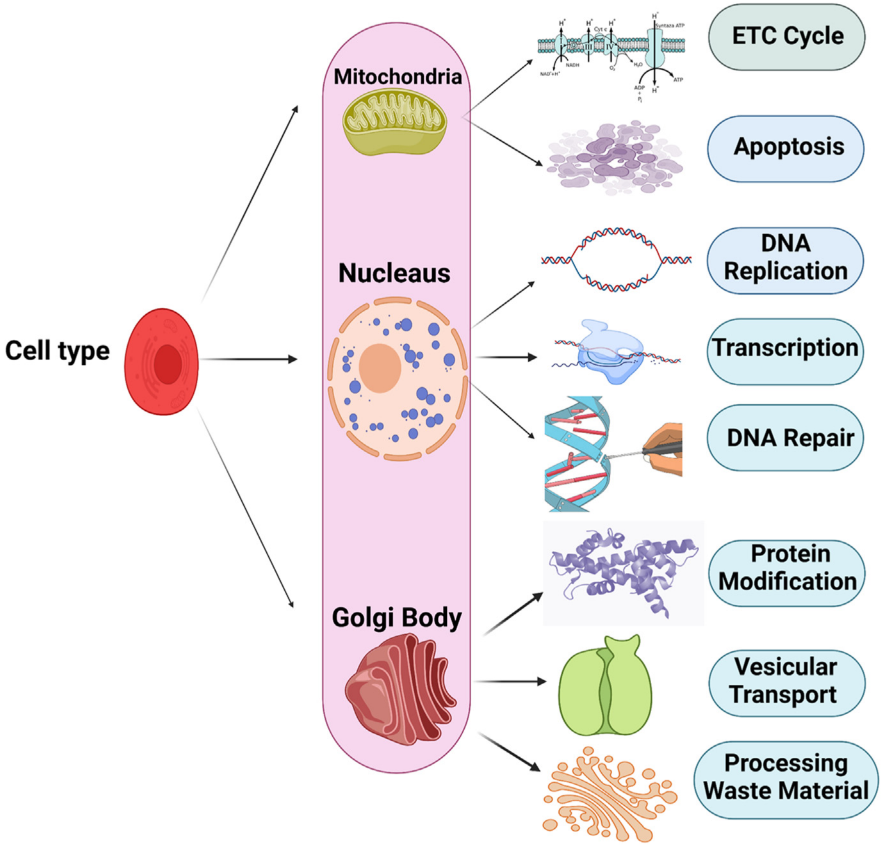

FIGURE 2.2

Hierarchical database model example.

connections and interactions are made. Because the data is more interconnected, accessing it is easier and takes less time.identifies data linkages on a many-to-many scale. Prior to the introduction of relational databases, this was the most commonly used model (Fig. 2.3).

2.3.4 Entity relationship model

In this information base paradigm, links are established by first breaking down the item of interest into its constituent parts, or substances, and then breaking down its attributes, or ascribes. Using connections, several different chemicals are linked together. It is acceptable to design an information base using this model, and that plan would subsequently be able to be converted into tables for use in the social model.In this information base paradigm, links are established by first breaking down the item of interest into its constituent parts, or substances, and then breaking

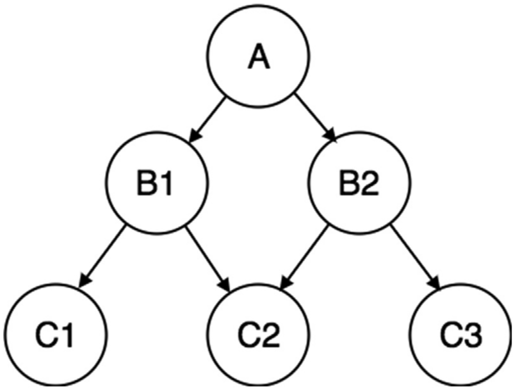

FIGURE 2.3

Network database model example.

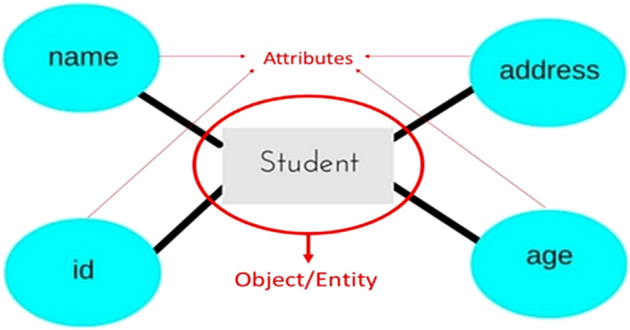

FIGURE 2.4

Entity relationship model example.

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2.3.5 Relational database model

The relational database model is now the most well-known information storage type that is accessible. The data in this model are set up in two-dimensional tables, and a common field is used to keep the link between the tables.Two-dimensional tables serve as the primary organizational framework for the information included in this model (basic flat file database). All of the information that may be categorized under a certain heading is saved in the appropriate column of that table. After that, relations are constructed between the several tables, hence the name.The relational database model is now the most well-known information storage type that is accessible. The data in this model are set up in two-dimensional tables, and a common field is used to keep the link between the tables.Two-dimensional tables serve as the primary organizational framework for the information included in this model (basic flat file database). All of the information that may be categorized under a certain heading is saved in the appropriate column of that table. After that, relations are constructed between the several tables, hence the name.

A relational database would be incomplete without a primary key and a foreign key.

- Primary key: It is the identifier column's purpose to store values that are one-of-a-kind (there is no room for duplication), and it is also the column that is used to describe a specific record inside a table in such a way that there is no overlap in the information stored. An example of this might be an accession number, index number, or any of the like.
- Foreign key: It is the column that connects one table to another table that does this (the primary key can also be made the foreign key). Relational databases are distinguished by this one-of-a-kind characteristic.

In 1970, E.F. Codd developed the relational model as a way to make information base management frameworks more independent of a particular application. This was done via the use of a relational database. It is a numerical model that has been defined in terms of predicate reasoning and set hypothesis, and different computer frameworks have used different implementations of it. Centralized server, midrange, and microcomputer frameworks have all made use of it. In reality, the things that are often referred to as social information bases put into action a model that is an estimate of the numerical model that Codd outlined. Relations, attributes, and regions are the three fundamental concepts that are applied extensively in social information base models. A link may be seen as a table consisting of segments and columns (Fig. 2.5).

2.3.6 Other models

The following is a list of some more models that are not as often used but are applied in certain circumstances:

- **Inverted File Model:** The information itself is used as a key in a query table, and the characteristics of the table are pointers to the location of each instance of a specific content item. The query table is used to organize the information.
- **Dimensional Model:** The dimensional model is a special kind of social model that is used to communicate with the data stored in information distribution centers in such a way that the data may be easily summarized by making use of online scientific handling or OLAP queries.
- **Graph Model:** Graph information bases make it possible to have a far more open structure than an organization data set does; each hub may be connected to another hub.

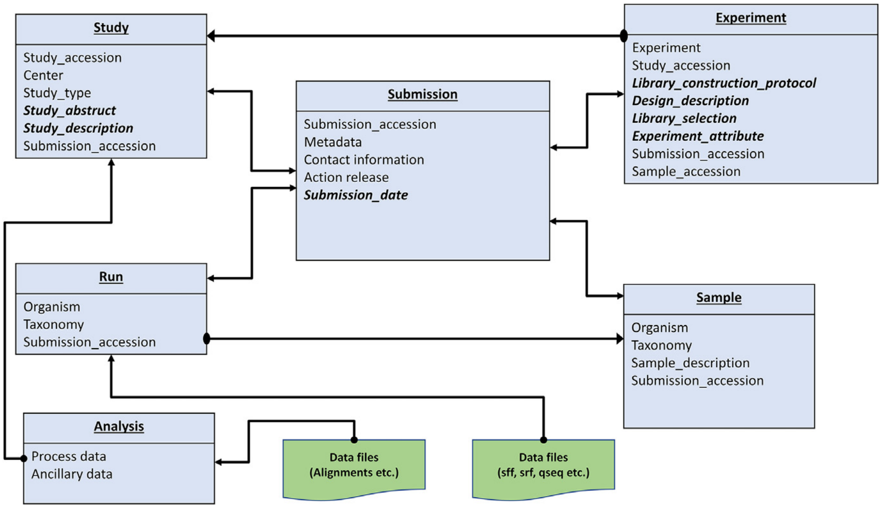

FIGURE 2.5

Relational database explained through the example of SRA (sequence read archive, repository of HTS sequence data available on NCBI).

- **Multivalue Model:** They can store the same path as relational databases, but in addition, they grant a degree of profundity that the relational model can only approximate by using sub-tables. This makes multivalue information bases "knotty" information in the sense that they can store the same path as relational databases.

2.4 Primary nucleic acid databases

The nucleotide database is a collection of different categories derived from a few different sources, such as GenBank, RefSeq, and TPA. The information about the genome, its quality, and the record layout provides the foundation for both biological research and the discovery of new knowledge. Primary nucleotide databases are a kind of biological database that holds data on nucleotide sequences that were obtained directly from researchers doing experiments in university labs, independent laboratories, or sequence centers. GenBank, EMBL, and DDBJ are the three most important databases in terms of the storage and accessibility of crude nucleic acid sequences to both the general public and analysts specifically. They are the repository of all raw nucleotide sequences, which is why they are considered to be the most important nucleotide sequence databases. The National Institute of Health at United States of America hosts the GenBank data repository, which may be accessed through the NCBI website. The DNA Databank of Japan (DDJB) is located in Japan, whereas the European Molecular Biology Laboratory (EMBL) is located in Europe.

To establish the highest level of synchronization possible across the three, each of them will accept submissions of nucleotide sequences, after which they will share fresh and updated data daily. Because they include the original sequencing data, these three databases are considered primary databases.

2.4.1 EMBL

Nucleotide Sequence Collection of the European Molecular Biology Laboratory (EMBL) is a comprehensive collection of primary nucleotide sequences that are maintained by the European Bioinformatics Institute (EBI). There are many different sources of data, some of which include individual scientists, gene sequencing facilities, and patent offices.

2.4.2 GenBank

The GenBank nucleotide sequence database is a collection of all of the publicly available nucleotide sequences and the protein interpretations of those sequences. This database is open to the public and contains annotations. As a crucial part of the NCBI, it is responsible for supplying and maintaining this data collection (INSDC). Get access to the DNA sequences of base pairs that were produced in laboratories all around the globe based on the characteristics of more than 100,000 different species. GenBank has developed into a valuable resource for scientists doing studies in natural environments. This database is now expanding at an exponential pace, as it has done ever since it was first created. The rate at which it is filling up is also exponential.

2.4.3 DDBJ

Its actual location may be found at Japan's National Institute of Genetics (NIG), which is located in the prefecture of Shizuoka. The DDBJ is the only nucleotide sequencing database that is specifically intended for use in Asia. Although Japanese researchers make up the bulk of DDBJ's data pool, the organization also welcomes contributions of sequences from researchers and donors from other nations.

2.5 Primary protein databases

The information included in biological databases may be extensively structured into data sets according to sequence and structure. The succession information bases are important for both the protein groupings and the nucleic acid arrangements, but the structure information bases are only relevant to the proteins themselves. After the insulin protein sequence was made available to the public in 1956, the primary knowledge base was developed in a very short amount of time. Insulin, which was not considered to be the most important protein to sequence, really is. Insulin's structure is shown by just 51 deposits, which are functionally analogous to letter sets in a sentence. These deposits form the sequence.

Around the middle of the 1960s, a nucleic corrosive cluster of yeast tRNA with 77 bases, which are single units of nucleic acids, was discovered for the first time. During this period, research was conducted into the three-dimensional structure of proteins, and in 1972, the now-famous Protein Data Bank was formed as the primary information source on protein structure. Back then, there were just 10 structures in all and currently it has more than 2,00,000 structures diposited.

While the basic information about how proteins are put together was kept at each research facility, in 1986 work began on putting all of this information into one place. This set of information is called the SWISS-PROT protein grouping data set.This data set currently has approximately 70,000 protein sequences from over 5000 model creatures, which is a small portion of every known life form.

Both academic institutions and businesses now have access to these enormous databases containing a wide variety of information sources, making it possible for them to conduct research and investigate the data. These are made available as open-access data in the greater context of the research network through the Internet. These information databases are regularly updated with the addition of new passages.

2.5.1 PDB

The Protein Data Bank, often known as PDB, is the only and most widely used library that includes both structural information and pictures of biological macromolecules. The Protein Data Bank (PDB) is the primary information source for many of the derived databases.It is where almost all of the structural bioinformatics studies that have ever been done begin (Joosten et al., 2011).

Files of biological molecules, along with their 3D coordinates, are what are largely kept in the PDB archive as the information that is saved there. These data detail the particles that make up each protein as well as their three-dimensional area in space. The aforementioned records may be accessed in a variety of formats (PDB, mmCIF, XML). The "header" section of a typical PDB-organized document is quite large and contains a lot of text. This section summarizes the proteins, reference data, and nuances of the structural arrangement. This section is followed by the succession and a not-insignificant rundown of the particles and their directions. The exploratory senses that are used to determine these atomic coordinates are also included in this file, which can be found in its entirety.

The RCSB Protein Data Bank adds to the data by making tools and resources for studying and teaching in the fields of molecular biology, structural biology, computational biology, and other related fields.

2.5.2 SWISS-PROT

The amino acid sequences that are utilized to connect amino acid sequences to information that is presently kept in the field of life sciences are retrieved from the SWISS-PROT protein database. An overview of the pertinent information is provided for each protein entry by combining the findings of the primary investigations with the characteristics estimated by simulations and other predictions, which may on occasion lead to contradictory conclusions. This is done to provide a comprehensive picture of the situation. This ends up producing an overview of the data that is applicable across disciplines. By establishing direct linkages to a variety of information databases, it is possible to get access to specialized knowledge that is not covered by SWISS-PROT. The annotation of human (the HPI project) and other model living creature passages is the primary emphasis of SWISS-PROT, but it does include explanatory elements for all species. This is done to ensure that sufficient annotation is accessible for individual agent proteins belonging to all protein families. As the High-quality Automated and Manual Annotation of Microbial Proteomes (HAMAP) project has done for species, some explanations may be applied to other families. This permits a more comprehensive understanding of the phenomenon to be gained. To stay current with the most recent findings in the field of logic, protein families and protein groups are subjected to ongoing analyses on a regular basis. By incorporating a growing amount of typically automated explanations, TrEMBL is working toward its eventual goal of covering all protein categories that are not yet handled in SWISS-PROT. This will be accomplished once all protein categories have been covered in SWISS-PROT (Boeckmann et al., 2003).

2.6 Secondary protein databases

A secondary database stores information that was either obtained from the original database or another secondary database. The primary database is the source of the information included in the secondary database. Information like the conserved sequence of a protein family, active site residues (derived from multiple sequence alignments of a group of proteins linked functionally or by sequence), and signature sequences are stored in a secondary database (identifying sequence). The Protein Data Bank (PDB) may be broken down into its parts, each of which is referred to as a secondary structural information base. Each alpha protein, each beta protein, and so on all have portions that are arranged in a certain order because of how the protein is structured. In addition, they provide information on the observed optional structural motifs of the protein. Components of the optional information base include SCOP, which was developed at Cambridge University; PROSITE, which was developed at the Swiss Institute of Bioinformatics; CATH, which was developed at University College London; and eMOTIF, which was developed at Stanford. Each of these parts was made by different analysts at their own research centers, where they are also being run.

2.6.1 CATH

CATH is a database that essentially holds a hierarchical categorization of domains of proteins based on how they fold. The domains that are stored in the Protein Data Bank (PDB) are recognized and categorized both manually and via the use of automated computer processes, and they are then placed into CATH. The CATH online interface allows for both simple searches for the categorization as well as thorough searches and downloads of the data (Sillitoe et al., 2019).

- **Class** is generated from the content of the secondary structure, and for more than 90% of the structures that have been saved for proteins, it has been automatically given.
- **Architecture** is a term that is used to define the fundamental alignment of secondary structures, apart from their connectivities (currently assigned manually).
- The **topology** level organizes the structures into groups based on how closely connected they are topologically and how many secondary structures they have.
- **Homologous superfamilies** are groups of proteins that have extremely similar activities and structures to one another.

2.6.2 SCOP

The purpose of the SCOP database is to provide descriptive and holistic details of the structural and evolutionary relationships between proteins whose three-dimensional structure is known and deposited in the Protein Data Bank. This information is sought after to better understand how proteins have evolved (Hubbard et al., 1999). The following are the primary levels of the classification:

- The term **"family"** refers to sets of proteins that are genetically and evolutionarily very closely connected. Current techniques of sequence comparison, including BLAST, PSI-BLAST, and HMMER, can identify their connection in the vast majority of instances.
- A **superfamily** is a group of protein domains that are only loosely connected. Their similarity is typically restricted to common structural characteristics, which, when combined with a conserved architecture of active or binding sites, or comparable processes of oligomerization, imply a likely evolutionary heritage. Their similarities are frequently limited to common structural aspects.

2.6.3 Prostate

Documentation values illustrating protein domains, families, and functional information are included in PROSITE, along with relevant examples and profiles that are used to differentiate between the different types.

PROSITE is complemented with a set of rules known as ProRule, which are reliant on profiles and examples. These rules enhance the ability of profiles and examples to skew results by providing additional information about amino acids that are either functionally or structurally important.

2.7 Composite sequence databases

The necessity to search through a variety of resources is eliminated thanks to the use of a composite sequence database, which combines a broad variety of important data sources. In their inquiry calculations, diverse composite information bases make use of unique fundamental data sets following a variety of principles. In addition to this, several search avenues have been streamlined and compiled for your convenience inside the comprehensive database. The National Center for Biotechnology Information (NCBI), which stands for "National Center for Biotechnology Information," keeps these nucleotide and protein databases on their huge, easily accessible, and redundant array of servers and gives researchers free access to them (Fig. 2.6).

2.7.1 Meta-databases

Metadatabases are knowledge bases that collect information about data sets to produce new data. This information is then used to generate new data. They could put together information from many different sources and present it in a new and more useful way, or they could focus on a certain disease or organism.

Examples are:

- An information discovery service was developed by the University of Antwerp and the Vlaams Instituut Voor Biotechnologie called BioGraph. It is based on the integration of more than 20 different types of databases.
- Information Framework for Neuroscience, hundreds of neuroscience-related resources are integrated into this system
- ConsensusPathDB is a database that integrates information from 12 different databases to provide a comprehensive view of molecular functional interactions.
- Entrez (National Center for Biotechnology Information)

2.8 Genomics and proteomics databases

Genetic information bases, which are also known as online genomic variation stores, may be provided for a single (locus-explicit) or several (generic) traits, or they may be provided expressly for a population or ethnic group (national or ethnic). In either case, the databases may be referred to as "online genomic variation stores." Both of these scenarios are not just possible, but likely. Genomic information bases are critical components of human genome informatics, which has grown exponentially in the postgenomic period as a consequence of a better understanding of the genetic etiology of human diseases and the visible confirmation of various genomic

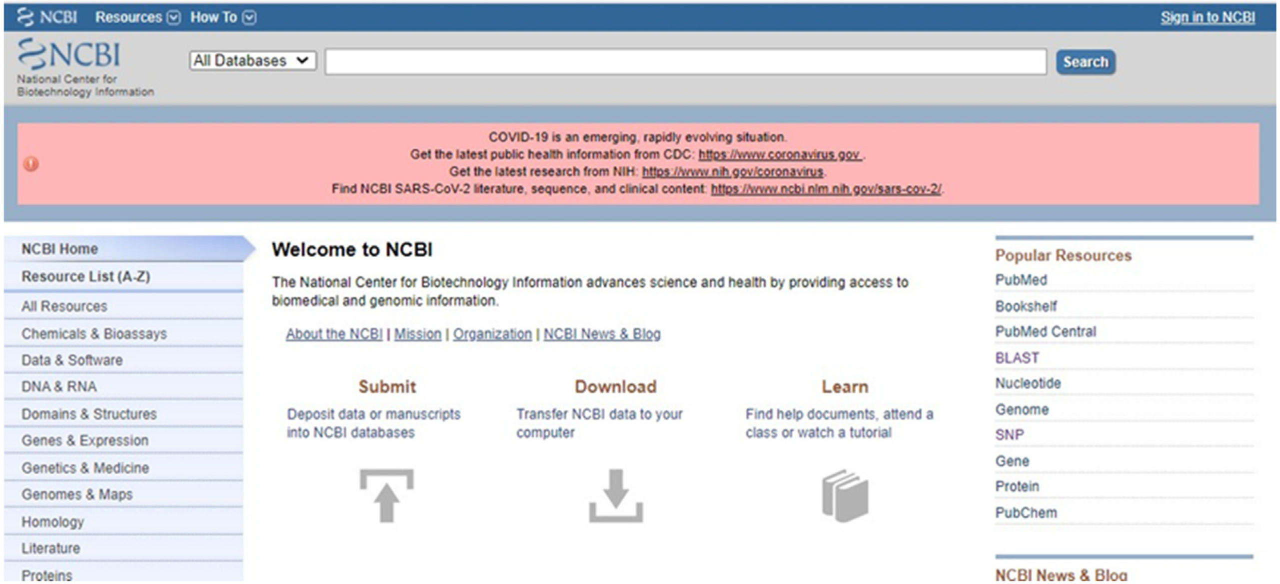

FIGURE 2.6

NCBI is one of the most popular composite databases among researchers hosted by NIH (National Institute of Health), this is the page loaded when we search ncbi.nlm.nih.gov.

variations. In the postgenomic period, the field of genomics informatics has experienced a period of explosive growth. This resulted in an improvement in our understanding of the genetic etiology of human illnesses as well as the visual confirmation of several different genomic polymorphisms. These resources organize the data and variations that were stated before with the hope that, in the future, they will be valuable not only for molecular diagnostics but also for doctors and analysts. To put it another way, the purpose of these resources is to reduce the amount of effort required for consumers to get the information that they need. The proteome databases are a collection of easily searchable, species-specific protein databases that merge publicly available sequence information with material that has been brought up to date via the careful curation of the scientific literature by trained professionals. Access to these databases is provided via the internet. As an example, the TUM serves as the database host for ProteomicsDB (Schmidt et al., 2018).

2.8.1 The search engines for literature

Internet access enables rapid access to a large quantity of clinical writing, including diaries, databases, word references, course readings, files, and electronic diaries. This, in turn, enables access to more variable, personalized, and effective instructional options. An online internet searcher is a piece of software or hardware designed to search for information on the World Wide Web. This information may take the form of website pages, photos, data, or other types of records. Web search tools for the web-based pursuit of clinical writing include Google, Google researcher, Yahoo internet searcher, and so forth, and information bases include MEDLINE, PubMed, MEDLARS, and so forth. Business web assets (Medscape, MedConnect, and MedicineNet) add to the rundown of asset information bases, giving a portion of their substance to open access. Some online libraries, such as the Medical Framework and the Emory Libraries, have been established as meta-destinations, providing important linkages to various wellness resources located all over the world. The availability of specific websites about dermatology, such as DermIs, DermNet, and Genamics Jornalseek, is a valuable addition to the list of electronic resources, which is always expanding. When searching for a certain category of information, a scientist has to keep in mind the benefits and drawbacks of the web crawler or data collection that they are using. In the field of medicine, information about the types of writing and levels of detail that are available, the user interface (UI), how easy it is to get started, how reliable the content is, and the time period that is covered makes it possible to get the most out of these resources.

2.9 Miscellaneous databases

2.9.1 Humans

Over the last 5 years, the Human Genome Project has had a significant amount of effect on the field of genetics research. This influence will soon be felt across the entire biological and medical professions. It won’t be long now. After describing them for centuries, we are now on the cusp of having a perfect comprehension of the cellular processes that take place within the human body. This is despite the fact that we have been describing them for millennia. Linkage analysis in families using limited sets of genetic markers was a technique that was necessary up until the late 1980s to establish the origin of hereditary disorders. This approach was time-consuming and required the use of genetic markers. At the end of that decade, a more all-encompassing strategy was proposed. It was called the Human Genome Project, and it included the mapping of all 80,000−100,000 of our genes as well as the decoding of our entire DNA sequence, which is 3 billion base pairs in length. This project was intended to be completed by the end of that decade. The first and most notable result of this program has been a spectacular acceleration in the process of determining the factors that contribute to inherited diseases. This initiative has considerably accelerated the development and distribution of advanced DNA technologies. Ten years have passed since the project's first conception. It has been shown that a malfunctioning gene or genes are to blame for the bulk of common genetic illnesses (150−200), in addition to a considerable number of rare genetic diseases (600−800). The majority of the disease-causing mutations have been uncovered, which has led to a huge improvement in the diagnostic capabilities of the condition. New genetic pathways have come to light as a result of these investigations, which are still ongoing. Genomic imprinting, expansion of triplet-repeat sequences, and defects in DNA repair are examples of some of these processes. This research, in turn, has led to breakthroughs in diagnostic techniques as well as the discovery of more disease-causing genes. The study of the so-called "genotype-phenotype correlation," which makes it possible for a more in-depth examination of the relationship between fundamental molecular flaws and functional disturbances of processes in cells, organs, and the organism as a whole, has also been made possible as a result of advancements made on a global scale. Because there are numerous stages in the chain of events that link cause and effect in the cell, as well as a complex web of interactions between various genes and other components of the environment, it is sometimes exceedingly difficult to establish these correlations. This is because genes and other elements of the environment interact in a complex web.

The Genome Database is a repository for data about human genes, clones, STSs, polymorphisms, and maps. This data may be accessed by the general public. The entries in the GDB are very well related to one another, to citations from the scientific literature, and to entries in other databases, including the sequence databases (Letovsky et al., 1998), OMIM, and the Mouse Genome Database. GDB is continuously receiving fresh mapping data from a variety of sources, including big genome centers as well as smaller mapping efforts. The database may be searched using a variety of methods, ranging from simple keyword searches to more involved queries. Over the last year, a significant number of brand-new capabilities have been included. For instance, Comprehensive Maps, which are integrated maps of the human genome and are now in the process of being created, are being used to facilitate positional searches and visual presentations. Printing maps and displaying ad hoc query results in a graphical style are two new features that have been added to the

GDB map viewer, often known as Mapview. The HUGO Nomenclature Committee is continuing to collaborate with GDB to maintain a record of the proposed and official gene symbols, in addition to the data associated with them. Because genome research is moving away from mapping and toward sequencing and functional analysis, the GDB schema is getting bigger.

2.9.2 Animals

Agriculture focused on animals will have to adapt swiftly if it is to be successful in meeting the food requirements of the future. The study of an organism's entire gene sequence is known as genomics, and it is a significant factor in the development of novel agricultural practises and technologies. Healthier animals that develop quicker, are less likely to become ill, and are better equipped to manage stressful or changing surroundings may be the result of using genomic information to improve how animals are bred. This may lead to the discovery of new animal breeds. Eliminating these issues would not only enhance the health of the general population but also cut expenses and losses for farmers. The satisfaction of clients may be increased by producing higher-quality goods. In addition, genomics may shed light on novel approaches to managing livestock production systems that are both more effective and friendlier to the natural environment. However, research into animal genomes is laborious, time-consuming, and costly. It might be too costly for individual researchers or smaller institutions to get the necessary tools and knowledge about genes. Because they do not know much about the latest technology, some scientists could be reluctant to employ them. Researchers will need to work together to make progress in animal genome research and find new ways to use what they know.

Experiments led to the discovery of the sizes of the genomes of over 6000 different animal species, and this information is compiled in a database called the Animal Genome Size Database. Each entry contains information on the taxonomy of the species, common names, how and from what tissues the size of the species' genome was estimated, and other relevant details. The database also includes connections to other locations on the internet where users may get images and further information about the species of their choice.

2.9.3 Fungi

To keep using fungi for the good of people, though, it's important to know how these organisms interact in both natural and artificial communities. People will soon be able to collect samples from different locations to investigate the possibility of complex fungus metagenomes. This will be an essential component of using fungus for the goals of industrial production, energy production, and climate management. However, unless we have well-defined reference data for fungal genomes, we won't be able to do a thorough analysis of these data.

An international research team is collaborating with the Joint Genome Institute of the Department of Energy on a project that will last for 5 years and will sequence the genomes of 1000 different types of fungi that are found across the Fungal Tree of Life. This will allow the team to learn more about the diversity of fungi. It is intended that the Fungal Tree of Life will be completed by sequencing at least two reference genomes from among the more than 500 families of fungi that are currently known. To do this, the main goal of this project is to collect data that can be used as a starting point for future studies on how plants and microbes interact, how microbes release and absorb greenhouse gases, and how metagenomes from the environment are sequenced.

A resource for functional genomics about pan-fungal genomes may be found inside the FungiDB database. It was put together by the Eukaryotic Pathogen Bioinformatics Resource Center, which contributed to its creation. Both the layout of FungiDB and its user interface are reminiscent of those seen in EuPathDB. This implies that complex and integrated searches may be done with a graphical method that is easy to comprehend. The most recent iteration of the FungiDB database contains the genomic sequences and annotations of 18 distinct fungus species that are representative of a wide range of fungal groups. The Basidiomycota orders Pucciniomycetes and Tremellomycetes, the Ascomycota classes Eurotiomycetes, Sordariomycetes, and Saccharomycetes, and the Mucormycotina lineage, which is the most fundamental "Zygomycete" lineage, are all included in this category. More information is available on the cell cycle microarray, the hyphal growth RNA sequence, and the yeast two-hybrid interaction in the FungiDB database.

2.9.4 Microorganisms

In our world, the group of creatures with just a single cell, known as microbes, is without a doubt the most abundant and diverse. There are approximately 12,000 known species, but there are likely millions more waiting to be discovered on our planet. Bacteria are capable of surviving practically every environment on Earth, including those that do not seem to be favorable to the existence of life. They have been seen as low as seven miles under the surface of the water and as high as 40 miles in the sky. There are a great number of types of bacteria that can survive in extreme environments such as intense heat, cold, and salt. The genomic sequences of microorganisms give a wide diversity of strains with varying degrees of quality and sampling intensity. Several significant diseases affect humans among them, as well as species that are interesting for reasons other than those related to medicine, such as biodiversity, epidemiology, and ecology. We have learned a great deal about evolution, microbial biology and ecology, and the existence of microbes from studying many different kinds of microbes, such as obligate intracellular parasites, symbionts, free-living bacteria, hyperthermophiles, psychrophiles, and aquatic and terrestrial microorganisms. The archeal genome of *Candidatus Parvarchaeum acidiphilum* was found to range in size from 45 kb to the largest draught assembly of *Mastigocoleus testarum* and the largest whole genome (14.7 mb) of *Sorangiumcellulosum*, an alkaline-adapted epiphyte maker. This discovery was made through research on mine drainage metagenomes.

On a workstation, the MBGD system allows for comparative and analytical examination of completely sequenced microbial genomes. The primary objective of the MBGD method is to generate an orthologous gene categorization table. This is accomplished by using all-against-all similarity correlations that have previously been computed between genes located in different genomes. By providing a list of organisms and parameters, users of MBGD will be able to generate their very own classification tables thanks to the incorporation of an automatic classification algorithm within the software. When the user is interested in creatures that belong to the same taxonomic category, this function is extremely beneficial since it allows them to narrow their search. The categorization table that was created has been kept in the database, where it may be seen in conjunction with data from individual genomes and information about the degree to which individual genomes are similar to those of other organisms.

2.9.5 Plant and crop genomic database

PlantGDB is a database that contains information on the molecular sequences of all plant species that have had major attempts made to sequence them. EST sequences are organized in the database into groupings referred to as "contigs." These contigs are assumed to represent individual genes. Contigs are assigned labels and, in cases where it's feasible, connected to the sections of genomic DNA to which they belong. The individual components of the genome sequence are assembled using the same method. The website known as PlantGDB aims to provide a method for locating groups of genes that are present in all plant species or that are exclusive to certain plant species alone. This is accomplished by integrating a variety of different bioinformatics tools, each of which makes it simpler to predict genes and compare them across different species. You can see the genomes of different species using PlantGDB that have undergone large-scale genome sequencing initiatives. It does this by putting together all of the EST and cDNA evidence for the gene models that are already available.

2.9.6 Organelle database

The online interface for the relational database known as Organelle DB may be found at the following address: http://organelledb.lsi.umich.edu. It includes a rundown of the key protein complexes as well as the proteins that may be found in organelles. Since its launch in 2004, Organelle DB has had a 20% expansion in size. It now contains over 30,000 proteins from 138 different eukaryotic species. The information provided by Organelle DB includes the location of each protein in the cell, the main sequence of the protein, and, if it is available, an in-depth description of the function that the protein serves. Every entry in Organelle DB has been annotated with words taken from the controlled vocabulary that is maintained by the Gene Ontology collaboration. The facts associated with protein localization are intrinsically visual, and Organelle DB contains a massive collection of photos related to biology. It includes 1500 micrographs of yeast cells that have been labeled to show the proteins.

2.9.7 Pathway databases

Pathway databases enable the compilation of lists of proteins and their associated functions, as well as the integration of such lists into networks that depict the functioning of an organism. The Reactome Knowledgebase is used as an example to show how a reaction space is built from manually curated experimental data, how semi-automated extensions of these manual annotations can be used to infer annotations for a large portion of a species' proteins, and how networks of functional annotations can be used to infer pathway relationships between different proteins that have been linked to disease risk by genome-wide studies.

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